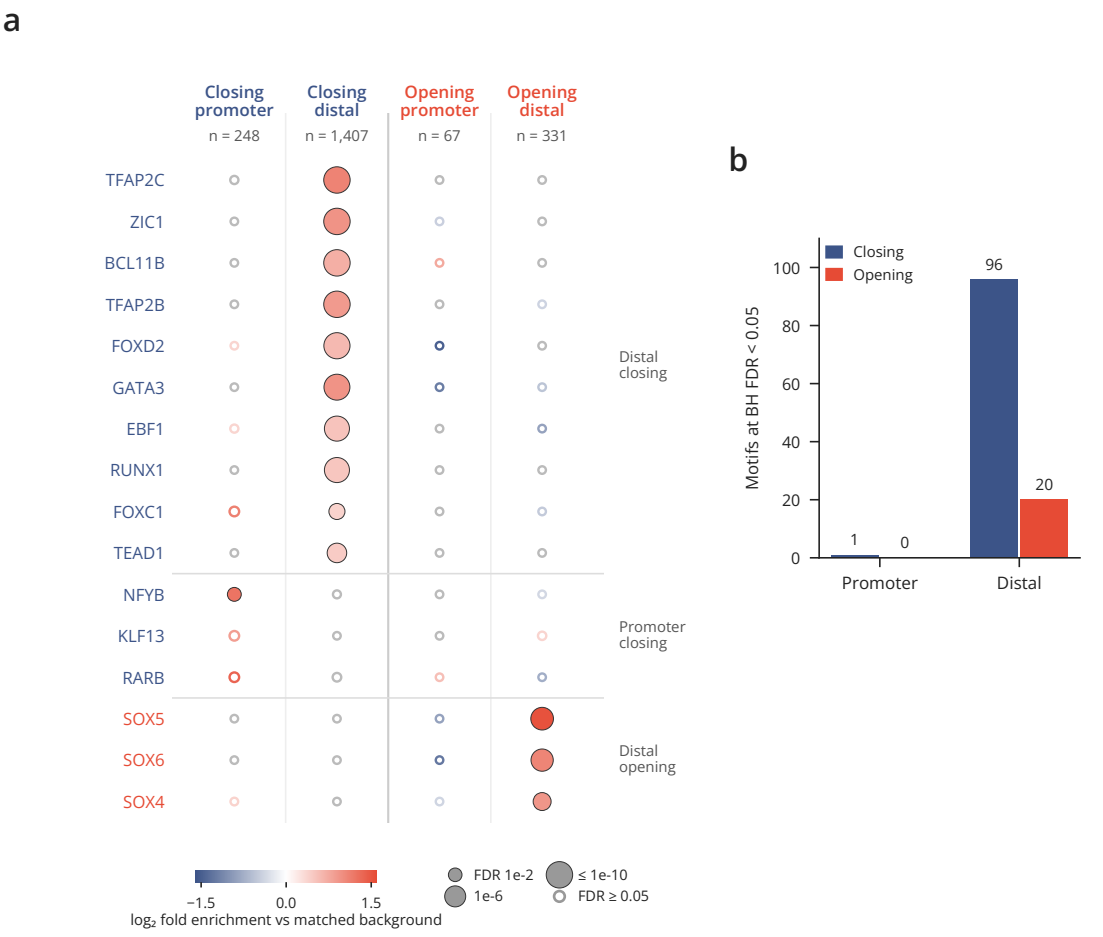
a
Closing promoter
n = 248
Closing distal
n = 1,407
Opening promoter
n = 67
Opening distal
n = 331
TFAP2C
ZIC1
BCL11B
TFAP2B
FOXD2
GATA3
EBF1
RUNX1
FOXC1
TEAD1
NFYB
KLF13
RARB
SOX5
SOX6
SOX4
Distal closing
Promoter closing
Distal opening
−1.5
0.0
1.5
log2 fold enrichment vs matched background
FDR 1e-2
1e-6
≤ 1e-10
FDR ≥ 0.05
b
Closing
Opening
Motifs at BH FDR < 0.05
100
80
60
40
20
0
96
20
1
0
Promoter
Distal